



HEADPHONES

Sounds good?* Don't buy it.

*the marketing jargon

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of the speaker that is baked into your earphones or headphones.

So, ideally, in the case of earphones with multiple drivers, the load of frequency reproduction is taken from one driver and distributed amongst three drivers. This means that each of the drivers is responsible for producing only one range of frequencies, which, when all put together, form a coherent audio signal that your brain interprets as a song or whatever. The process of making sure that one particular driver is producing a given range of frequencies is called tuning.

Tuning is a very intricate process which requires a lot of research and expertise. This means that if the drivers, even if there are many of them in a single earphone, are not tuned to reproduce individual frequency ranges, you might get a loud audio signal, but it may not be accurate.

The bottom line is that when buying earphones with multiple drivers, make sure that they are tuned well, and to check that, you will need to experience the device in person. When doing that, you can play a sound frequency sweep that covers frequencies between 20Hz and 20,000kHz, using the device you will use with the earphones and test whether the earphones are capable of accurately reproducing the entire audio spectrum or not.

Tuning matters

You'd have read in the last section of the article that the tuning of the drivers is important. But what does it mean?

So, as we said, tuning a headphone/earphone driver means the calibration of the driver to reproduce certain frequencies. This varies across manufacturers. One manufacturer might tune their drivers to produce lower frequencies better than the mids and the highs, making the sound bass heavy. The others might have tuned their drivers to give a V-shaped sound signature, where the lows and the highs are boosted, with the mids being lowered.

This shows that, unless tuned with precision and care, two drivers of the

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hen you're buying a pair of headphones, especially in the current landscape

where getting knocked over by the hype train is easier than ever, you need to be extremely vigilant. There are things that are actually not what they are made to look like. For example, Apple Music supports playing lossless audio using the ALAC CODEC, but Apple's own AirPods do not support that CODEC. While it is very hard for untrained ears to distinguish between certain types of audio transmissions, having them ready and eventually training yourself to be able to distinguish between them is nice to have. And, especially when high-end audio products like headphones and earphones are available in plenty, you'd not want to spend a fortune for something that you could get for half the price just because the pricier product was marketed to look fancier.

So, to help you map your way through the world of headphones and earphones, we have some tips and words of caution coming from our audio experts –

Drivers – the more, the merrier? Maybe not.

A lot of manufacturers advertise the presence of multiple drivers in their earphones. It may seem like the higher the number of drivers, the better the earphones would sound. However, when looking at it scientifically, it is not always the case.

Sound signatures are generally divided into three broad frequency Ranges – lows, mids, and highs. Each of the individual frequency bands have their own frequency values. In the simplest words, these denote the number of vibrations that are there in the waveform of the sound wave produced at that frequency. These vibrations are produced by the diaphragm